Database Backup Recovery

Overview

Having a database backup recovery plan is important for protecting valuable data, ensuring business continuity, and maintaining trust in your organization's ability to manage information securely. In this example, three models will represent the schedule of backups at different levels of intensity. A flow chart will be used as a visual to show the recovery process steps. In the end, there will be a turnover process and questions to evaluate the backup and recovery process.

Backup Schedule Models-

For backups to be secure, backups will be encrypted and stored in a locked area only to be accessed by authorized personnel. After every week, backups should be deleted for a daily backup schedule. Backups for a weekly backup schedule should be deleted every two months. There will be a log for who stores and deletes backups to keep a record in the case of a security breach or problem.

Strong Model-

STRONG	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
12:00AM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
1:00AM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
2:00AM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
3:00AM	Full Backup	Full Backup	Full Backup	Full Backup	Full Backup	Full Backup	Full Backup
4:00AM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
5:00AM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
6:00AM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
7:00AM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
8:00AM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
9:00AM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
10:00AM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
11:00AM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
12:00PM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
1:00PM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
2:00PM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
3:00PM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
4:00PM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
5:00PM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
6:00PM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
7:00PM	Full Backup	Full Backup	Full Backup	Full Backup	Full Backup	Full Backup	Full Backup
8:00PM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
9:00PM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
10:00PM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
11:00PM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log

This strong model has two full backups within the same day and the remaining hours are transaction logs capturing every transaction, detailing the actions taken, such as inserts, updates, and deletes, along with timestamps and other relevant metadata.

Intermediate Model-

INTERMEDIATE	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
12:00AM							
1:00AM							
2:00AM							
3:00AM	Full Backup	Full Backup	Full Backup	Full Backup	Full Backup	Full Backup	Full Backup
4:00AM							
5:00AM							
6:00AM							
7:00AM							
8:00AM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
9:00AM							
10:00AM							
11:00AM							
12:00PM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
1:00PM							
2:00PM							
3:00PM							
4:00PM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
5:00PM							
6:00PM							
7:00PM		Full Backup	Full Backup	Full Backup	Full Backup	Full Backup	
8:00PM							
9:00PM							
10:00PM							
11:00PM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log

The intermediate model is less complex with four transaction logs each day, two full backups each day during the week, and only one full backup each day of the weekend.

Simple Model-

SIMPLE	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
12:00AM	Full Backup				Full Backup		
1:00AM							
2:00AM							
3:00AM							
4:00AM							
5:00AM							
6:00AM							
7:00AM							
8:00AM							
9:00AM							
10:00AM							
11:00AM							
12:00PM							
1:00PM							
2:00PM							
3:00PM							
4:00PM							
5:00PM							
6:00PM	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log	Transaction Log
7:00PM							
8:00PM							
9:00PM							
10:00PM							
11:00PM							

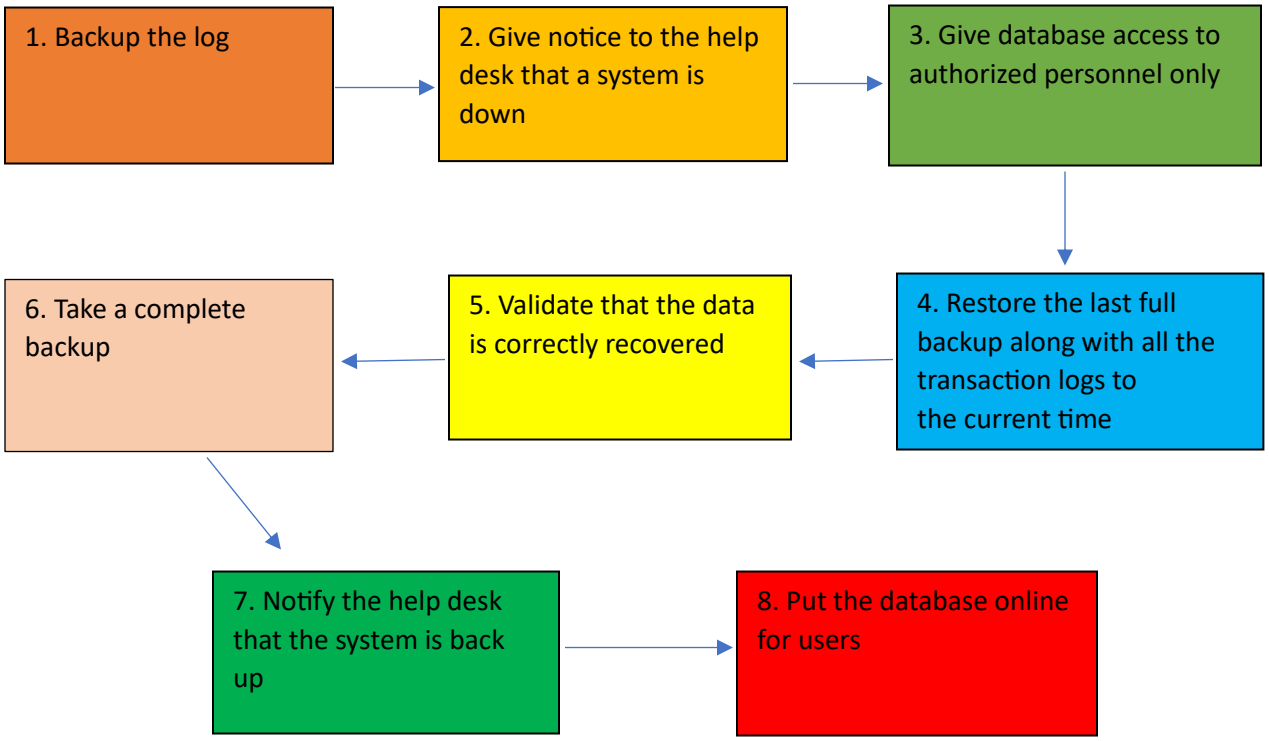
The Simple Model has two full backups per week and one daily transaction log. These simple models would be used in databases that don't handle large volumes of transactions or complex data structures.

Support Protocols-

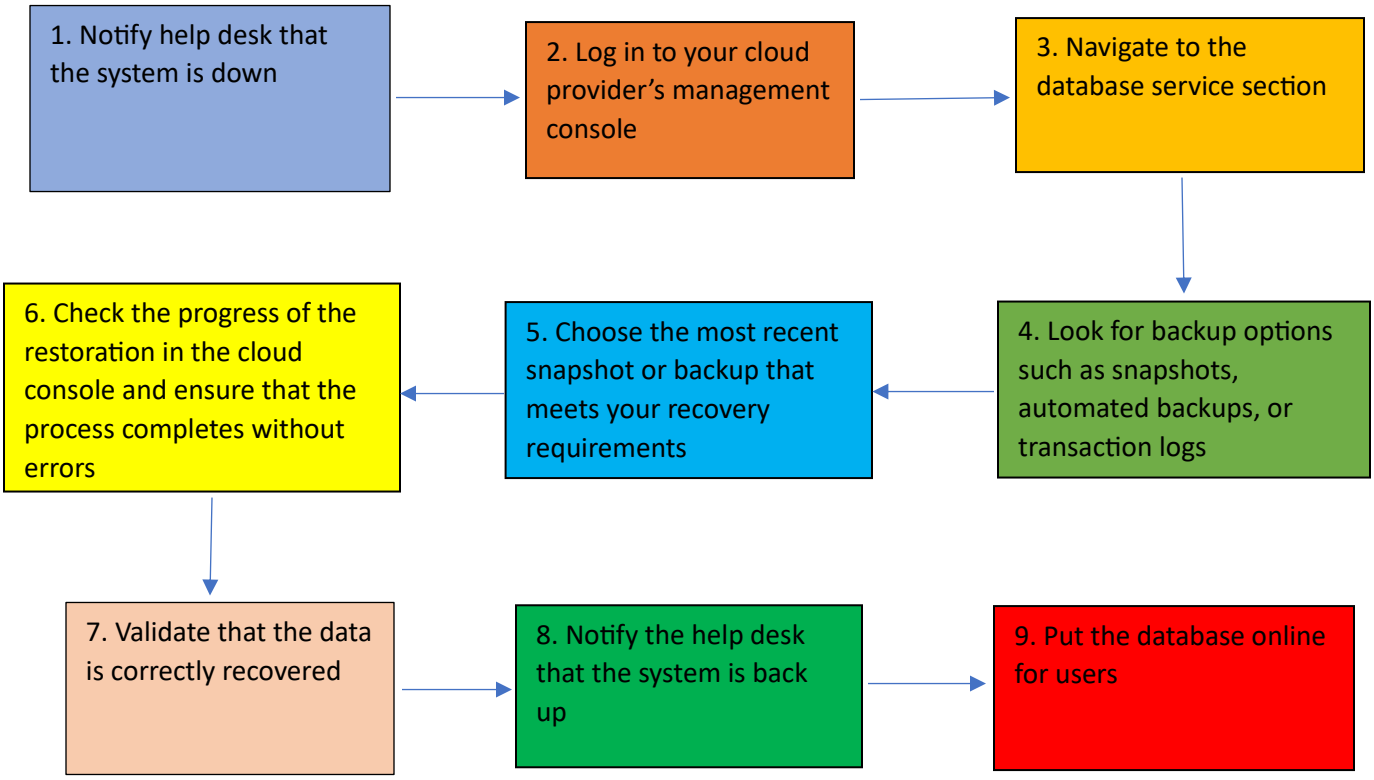
If issues were to occur at any time you should contact the help desk for support. For emergencies, you may contact the manager through their emergency phone number. Be sure to prioritize systems according to their importance to the company. This means if you are working on an error with a testing environment system and a financial transaction system fails, you MUST transition your energy to the financial transaction system. In case you are unable to tend to your usual department, you should leave detailed notes or comments on the current processes of your systems so that someone from another department can pick up from where you left off.

Recovery Process-

This flowchart represents the steps to take when a recovery is needed for a local platformed system.



This flowchart represents the steps to take when a recovery is needed for a cloud platformed system.



Metrics-

- What was the backup success rate?
- How long did it take to complete the backup?
- What was the total downtime for users?
- Did the recovery meet the defined RTO and RPO?
- Were there any issues with the recovery process?
- What did this cost the company?
- Did we communicate effectively during the recovery process?
- What was the error rate during the recovery?
- User satisfaction score post-recovery
- What/who caused the error in the database?
- How frequently does the system go down?

Turnover Control

The two types of turnover controls are simple and complex. Simple turnover controls are basic, straightforward measures designed to monitor and control transactions or data flow with minimal complexity. Complex turnover controls involve intricate systems and processes that utilize advanced technology, automation, and data analytics to monitor and manage transactions and data flows. Scheduling simple turnover controls is easier than complex ones because they have a low chance of causing negative impacts on the systems. On the other hand, complex turnovers must be overviewed more than simple turnovers since they have a higher chance of causing negative impacts on the systems. Complex turnovers are done at a time when there is low activity of users so that if something were to go wrong, it wouldn't impact lots of people such as performing complex controls at two in the morning.

Below is a template for what a turnover control form would look like:

Turnover/Change Control Form

Form ID: _____
Date: _____

1. Change Request Details

- **Change Title**:

- **Requestor Name**:

- **Department**:

- **Contact Information**:
  - Email: _____
  - Phone: _____

2. Description of Change

- **Description**:

- **Reason for Change**:

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3. Impact Assessment

- **Systems/Processes Affected**:

- **Expected Impact on Users**:

- **Risk Assessment**:

4. Change Implementation Plan

- **Proposed Implementation Date**:

- **Implementation Steps**:
  1.
  2.
  3.
- **Rollback Plan**:

5. Approvals

- **Change Requestor Signature**:
  _____ Date: _____
- **Manager/Supervisor Approval**:
  _____ Date: _____
- **IT/Development Team Approval**:
  _____ Date: _____
- **Compliance/Quality Assurance Approval**:
  _____ Date: _____

6. Post-Implementation Review

- **Date of Implementation**:

- **Outcome of Change**:

- **Post-Implementation Issues**:

- **Lessons Learned**:

Turnover Procedure:

Simple: The simple turnover control procedure begins with a team member identifying a need for change and completing a Turnover/Change Control Form, detailing the change's title, description, reason, and affected systems. The completed form is submitted to the immediate supervisor for review and approval. Upon approval, the designated implementation team prepares for the change, communicates the plan to affected users, and executes the change while monitoring progress. After implementation, the team validates the change's success, gathers user feedback, and documents the outcome in the form, noting any lessons learned. This documentation is then stored for future reference, ensuring a systematic approach to managing changes while minimizing risks.

Complex: The complex turnover control procedure begins with a detailed change request initiated by a team member, who completes a comprehensive Turnover/Change Control Form outlining the change's scope, impact, and risk assessment. This request is then submitted for multi-level approval, including assessments from management, IT, and compliance teams. Once approved, a cross-functional implementation team develops a detailed implementation plan, incorporating automated workflows and testing protocols. Before execution, stakeholders are informed of the planned change, its timeline, and potential impacts. The implementation is conducted with real-time monitoring and immediate issue resolution. Afterward, a thorough post-implementation review is conducted to validate data integrity, gather feedback, and analyze outcomes against predefined metrics. Finally, all documentation, including lessons learned and compliance records, is archived for future reference and continuous improvement of the turnover process. For complex turnovers, a full backup is required before and after the change in case the change causes system failures.